

CRITICAL REASONING 9-9-26

Directions for questions 1 to 15: Select the correct alternative from the given choices.

The conclusion of a study from Finland, a land not noted for its joie de vivre, published last week in BMC Public Health, found that pessimism was associated with an increased risk of death from coronary heart disease. The researchers also looked at Optimism in the same group of middle-aged Finns and failed to find any association with a decreased risk of coronary heart disease.

Which of the following, if true, most helps to resolve the apparent paradox?

- ☐ a) People learn to be positive or negative in childhood, depending on whether they were raised in families where the parents were of a high or low socioeconomic status.
- ☒ b) Optimism and pessimism are two attitudes; they are not opposite ends of the same emotional spectrum.
(Correct Answer - ✓)
- ☐ c) People who expect the best things in life actively take steps to promote health.
- ☐ d) The immune response of an individual waxed and waned depending on his/her optimism or pessimism.

Solution:

A study from Finland, a land not noted for merriment, concluded that pessimism was associated with an increased risk of death from coronary heart disease. But they could not find any association between optimism and decreased risk of coronary heart disease. What information can help explain this apparent paradoxical situation? If we take optimism and pessimism as just two attitudes and not as opposite sides of our emotional spectrum, the paradox can be resolved.

- (A) How a person develops either of these attitudes doesn't explain the paradox.
- (B) CORRECT. This statement properly explains why no association between optimism and decreased risk of coronary heart disease was noticed while association between pessimism and increased risk of coronary heart was evident.
- (C) This statement is irrelevant to explaining the paradox; it doesn't explain it.
- (D) Immune response of an individual has not been discussed in the paragraph.

Choice (B)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

According to a new research, Americans without a college degree are becoming more likely to die in middle age, reversing decades of progress toward better health. The researchers found evidence that trade with China has resulted in greater rates of suicide and poisoning including fatal drug overdoses after 2000. People's preference for cheap Chinese goods forced a number of U.S. factories to close, and many of the Americans who were laid off fell into depression or addiction. The researchers conclude that, for workers, certain changes in the US economic policy proved fatal.

The conclusion above is based on which of the following assumptions?

- ☐ a) Americans are unable to withstand stress.
- ☒ b) America has allowed more Chinese imports since 2000.
- ☐ c) Trade with China has resulted in hazardous health effects.
- ☐ d) A college degree is a passport to a decent job in America.

(Correct Answer - ✓)

Solution:

Americans without a college degree have become more likely to die in middle age following America's liberalization of trade with China. People's preference for Chinese goods has resulted in the closure of many factories, and many Americans, who were laid off fell into depression or addiction. A research concludes that certain changes in the American economy were fatal to workers. What assumption does the argument make? The argument states that trade with China and closing of US factories followed by laying off of workers resulted in greater rates of suicide and poisoning. The conclusion is that for workers, certain changes in the economic policy were fatal. As the stated premises are inadequate to support the conclusion, it is reasonable to assume that since 2000, the US has allowed import of Chinese goods affecting demand for goods produced in the US.

- (A) Though this option might be consistent with the argument, it is not assumed as it is not applicable to all Americans.
- (B) CORRECT. This is the best choice for the assumption. This assumption underlies the argument.
- (C) This is stated in the conclusion, so it cannot be an assumption.
- (D) The conclusion doesn't suggest any link between a college degree and job security.

Choice (B)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

Globally there has been an increased intake of fatty foods. We are well aware that eating fatty foods leads to weight gain and obesity. Obesity is responsible for multiple health conditions, psychological and social problems. It could lead to joint problems, increased risk of osteoporosis, cancer, urological and reproductive problems, diabetes and coronary heart disease. In fact, it could triple the risk of heart disease. Hence, it is advisable to avoid fatty foods in toto.

Which of the following, if true, most weakens the argument above?

- ☐ a) Excess fat in the body can be reduced considerably by exercising regularly.
- ☐ b) Fats are present naturally in many foods.
- ☒ c) Fat is one of the three micronutrients along with proteins and carbohydrates that help maintain good metabolism. (Correct Answer - ✓)
- ☐ d) Obesity and many non-communicable diseases can be prevented by eating fruits and vegetables, legumes and whole grains and nuts and by engaging in regular physical activity.

Solution:

There has been an increase in the intake of fatty foods all over the world. Fatty foods lead to obesity which is responsible for many health conditions including heart disease. Hence fatty foods should be totally avoided. Which information weakens the argument that fatty foods should be totally avoided? The conclusion that fatty foods should be avoided in toto is made on the grounds that they are bad for health. If there is something in fatty foods that has some health benefit, the argument can be weakened.

- (A) This statement talks only about reducing excess fat 'considerably' by exercising regularly, and so it doesn't in any way support the conclusion.
- (B) Presence of food naturally in foods is not related to the argument.
- (C) CORRECT. If fat helps metabolism, which is necessary for life itself, fat should be taken.
- (D) This statement is not related to the consumption of fat, so this is irrelevant.

Choice (C)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

A toll road was built between two cities of two neighbouring States. The two State governments, which awarded the contract to a particular company allowed a return of 20 per cent to the company for 30 years, which was the duration of the contract. After 16 years of toll collection, the High Court of one of the States, heeded to public interest litigation and ordered the company to stop toll collection, though 14 years were left.

Which of the following can be logically concluded from the above?

- ☒ a) The contract between the two States and the company is not inviolable. (Correct Answer - ✓)
- ☐ b) Companies may become wary of investing in toll roads in any State in future.
- ☐ c) Motorists can use other toll free roads connecting the two cities.
- ☐ d) Fixing toll is not at the discretion of the State government and the company that constructs the road.

Solution:

After 16 years of toll collection, a company that had constructed a toll road was asked to stop toll collection by the High Court of one of the States, resulting in under recovery of debt. What conclusion can be reached about the verdict of the High Court? The passage states that the Court asked the company to discontinue toll after 16 years though originally the contract was for 30 years. Thus, the Court did not maintain the original contract between the company and the two States. It is logical to conclude that the contract between the two states and the company is not sacred in the eyes of the Court.

- (A) CORRECT. This statement is a conclusion that can be logically concluded from the argument.
 - (B) Nothing in the passage points to other companies or other states, so this cannot be concluded.
 - (C) The passage provides no information about toll free roads, so this option is irrelevant.
 - (D) The argument is not about the toll rate; it is about toll collection, so this is also not a logical conclusion.
- Choice (A)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

The government's decision on November 8 to immediately ban Rs.500 and Rs.1,000 notes that account for 86% of all currency in circulation has impacted a raft of sectors. The decision has caused a sharp drop in demand for goods and services. While farmers and small industries will bear the brunt and sectors like transport and real estate will visibly be in pain, several other industrial sectors will have to scale back services or production. Hence, growth will be hit across all sectors.

Which of the following, if true, most strengthens the argument?

- ☐ a) The decision will eventually lead to a world of cash free transactions where coins and notes are replaced by bits and bytes and spending, earning and investing get updated in real time via smart phones.
- ☐ b) Cash is unlikely to be back as the preferred option for transactions in future.
- ☐ c) Even after demonetization, people are buying all essentials.
- ☒ d) Consumers have turned frugal with demonetization. (Correct Answer - ✓)

Solution:

The government's decision to ban ₹500 and ₹1000 notes has caused a sharp drop in demand for goods and services. The argument concludes that growth will be hit across all sectors. Which of the options, if true, would most strengthen the argument? An option that can most contribute to a fall in demand for goods can be the right option.

(A) The world becoming one of cash free transactions is irrelevant to fall in demand.

(B) This option too can be eliminated on the above grounds.

(C) Since this option is restricted to buying of essentials, it will not affect the conclusion much.

(D) CORRECT. This shows that demand will drop as the buying habit of people has changed.

Choice (D)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

A study conducted by researchers at the University of California at San Diego School of Medicine and the Skaggs School of Pharmacy and Pharmaceutical Sciences, used chemical analysis from molecules on phones to determine basic lifestyle choices of the owner. Personal routine clues such as diet, cosmetic or makeup use, clothing, and medication could be gleaned from the wide-reaching analysis, as well as environmental clues such as ocean and sunscreen molecules that could point toward locations where the owner recently visited. The study concluded that a simple swab from the surface of a phone can accurately predict a person's lifestyle choices.

Each of the following, if true, can contribute to the accuracy of the prediction of a person's lifestyle choices EXCEPT:

- ☐ a) The molecules on phone surfaces linger for extended periods of time.
- ☐ b) People spend a lot of time on phones.
- ☒ c) Every now and then, new organic molecules are being created on the surface of phones by reactions with others. (Correct Answer - ✓)
- ☐ d) The outermost layer of human skin carries chemical components which reflect the body's inner chemistry.

Solution:

A study says that a simple swab from a person's phone can accurately predict his lifestyle choices such as diet, make up use, clothing, medication etc. Which of the options does not help in the accurate prediction of lifestyle choices?

- (A) This factor can make the prediction accurate since the molecules will continue to give clues to the choices the person has.
- (B) If people spend a lot time on phones chances are that traces of his choices will be there on the phone.
- (C) CORRECT. If the molecules on the surface of the phone undergo constant changes due to bonding with other molecules, accuracy of prediction will be adversely affected.
- (D) If this is true, the diet the person takes can be accurately predicted.

Choice (C)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

The Indian Finance Minister has said that the demonetization of Rs.500 and Rs.1000 notes will take the country further towards digital currency. Further, the cancellation of notes, which accounted for about 86% of the value of currency, will tackle black money, fakes, corruption, and terror financing, the FM has added.

If the statements above are true, which of the following can properly be inferred on the basis of them?

- ☐ a) The volume of trade is likely to grow drastically in future.
- ☐ b) Following demonetization, the growth of the economy is unlikely to be hampered.
- ☐ c) Rs.500 and Rs.1000 notes will not be printed in the immediate future.

☐ d) Currency in circulation will be less than it was during the pre demonetization days. (Correct Answer - ✓)

Solution:

Demonetization of ₹500 and ₹1000 notes will lead to digital currency and tackle black money, fakes, corruption and terror fighting, says the FM. What inference can be drawn from the above? The argument implies that huge quantity of high denomination notes in circulation and the absence of digital currency were responsible for evils like corruption, black money etc. Hence it is logical to infer that there will be fewer notes in circulation.

- (A) The passage makes no reference to the volume of trade, hence this option is irrelevant.
- (B) This is also beside the point.
- (C) This may be a close option, but it is the value of currency in circulation, not the denomination that matters.
- (D) CORRECT. Based on the given information, this is a logical inference.

Choice (D)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

Leading brokerages, Credit Suisse and CLSA, have forecast a weaker rupee going ahead after the Indian unit hit a record low recently on rising expectations of an interest rate increase by the US Fed this month. Credit Suisse technical analysts expect rupee to break the key level of 70 to a dollar and hit 72 as higher interest rates in the US will lead to rupee falling. In this situation, it is advisable to buy stocks of IT and pharmaceutical companies.

The argument above logically depends on which of the following assumptions?

- ☐ a) Interest rate increase in the US does not affect IT and pharmaceutical companies.
- ☐ b) IT and pharmaceutical companies do not import raw materials or ingredients from the US.

☐ c) IT and pharmaceutical companies receive most of their revenues in dollars. (Correct Answer - ✓)

- ☐ d) Investors normally take advantage of a falling rupee by buying stocks of IT and pharmaceutical companies.

Directions for questions 1 to 15: Select the correct alternative from the given choices.

Researchers at the University of Montreal have studied the behavior of children who are exposed to tobacco smoke at home. The study concludes that children exposed to tobacco smoke at home are prone to antisocial behavior, aggression and are more likely to drop out of school.

Which of the following, if true, most strengthens the argument?

- ☐ a) Animal studies have found that exposure to tobacco smoke resulted in abnormal brain development.
- ☒ b) Animal studies have suggested that developing brain is most vulnerable to any toxic environment input.
(Correct Answer - ✓)
- ☐ c) Another study has showed that even second-hand tobacco smoke can harm the brain.
- ☐ d) A recent study on mice has showed that even transient exposure to tobacco smoke can affect brain development.

Solution:

A research has shown that children exposed to tobacco smoke at home are prone to antisocial behavior, aggression and are likely to drop out of school. Which option, if true, would most strengthen the argument? All options relate to the effect of tobacco or similar toxic substance on the brain. An option that points to a deleterious effect of such things on the brain can strengthen the conclusion.

- (A) This can be considered, but since the conclusion relates only to children, this option, which states an effect on all, may be eliminated.
- (B) CORRECT. If developing brain is vulnerable to any toxic environment input, the conclusion is strengthened.
- (C) This is inappropriate because the conclusion is about second hand tobacco smoke. Young children don't smoke.
- (D) The argument concerns only children, hence this need not be considered.

Choice (B)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

Though it is generally believed that proteins lurking in blood have rejuvenating effects, a group of scientists in California found high levels of certain proteins in the blood samples collected from people who had become prematurely old. According to these scientists, molecules in blood can shift the pace of ageing in various body tissues.

If the statements above are true, which of the following can properly be inferred on the basis of them?

☒ a) Resetting of abnormal levels of proteins in blood will slow down ageing process and prevent age related diseases. (Correct Answer - ✓)

☐ b) High levels of certain proteins can hamper the growth of bodily tissues.

☐ c) Any individual can be made a champion in swimming or running by tampering with his blood.

☐ d) Body tissues deteriorate with old age.

Solution:

Refuting the popular perception that proteins in blood have rejuvenating effect, scientists found high levels of certain proteins in the blood of some people who had become prematurely old. Scientists say that molecules in blood can alter the pace of aging in body tissues. What inference can be drawn from the information above? Since the passage mentions a connection between the presence of proteins in blood and ageing it is logical to infer that by controlling protein levels aging process can be slowed down.

(A) CORRECT. This is a possible inference that can be drawn from the information above.

(B) This can be a conclusion, not an inference.

(C) This is far-fetched.

(D) This is irrelevant to the argument

Choice (A)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

Scientists studied two bird flu viruses, H5N1, a type 1 branch, and H7N9, a type 2 branch, each of which has caused hundreds of cases of severe illness or death in humans. They are concerned that in the future either of these strains could gain mutations and trigger a deadly pandemic. They say that if a person's first flu infection belonged to the same branch as the avian strain, they had a 75 per cent reduced chance of hospitalisation and about an 85 per cent smaller chance of dying. They observe that H5N1 tends to cause far more mortalities in younger people while H7N9 tends to be most devastating in the elderly population.

Which of the following can be logically concluded from the passage above?

- ☐ a) Younger people are more likely to encounter birds infected with the virus.
- ☐ b) The virus is more aggressive in elderly people.
- ☐ c) Type 1 virus is more aggressive than type 2 virus.
- ☒ d) Six to seven decades ago, all bird flu viruses in circulation belonged to the type 1 branch. (Correct Answer - ✓)

Solution:

Scientists studied H5N1, a type 1 bird flu virus and H7N9, a type 2 bird flu virus. They say that mutations in these viruses will cause a deadly pandemic in the future. Further, they say that if a person's first flu infection belonged to the same branch as the avian strain, they had a 75 per cent reduced chance of suffering or death. H5N1 tends to cause far more mortalities in younger people while H7N9 tends to be most devastating in the elderly population. What conclusion can be reached about the two strains of viruses based on the passage? The passage states that H5N1 causes more mortality in younger people while H7N9 is most devastating in the elderly population. This is because young people have had no earlier history of H5N1 virus, a type 1 virus.

- (A) No information is available in the passage about who has more chances of contact with birds, the young or the old.
- (B) This doesn't match the findings of scientists; they say in case of H5N1, the old are less affected.
- (C) This is also irrelevant as both the types are aggressive in different age groups.
- (D) CORRECT. This conclusion can be drawn from the passage as H5N1, a type 1 virus, causes less damage to people who were born 6 or 7 decades ago.

Choice (D)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

Empowering women can be an economic game changer for any country. For instance, if women were to participate in the labour force to the same extent as men, national income could increase by five per cent in the US and nine per cent in Japan. In case of India, if all its working age women are employed, its national income will increase by 27 per cent.

The conclusion above is based on which of the following assumptions?

☐ a) Women in India are as capable as men in doing any work.

☐ b) Women in India seldom take up paid work.

☒ c) There are enough employment opportunities for women in India. (Correct Answer - ✓)

☐ d) All over the world, women are a disadvantaged lot.

Solution:

If all its working age women are employed, India's national income will increase by 27 per cent. What assumption does the argument make? The argument states that women taking up employment will be an economic game changer for any country, and particularly for India. It is reasonable to assume that they don't take up employment even though there are sufficient opportunities for them in India.

(A) The conclusion concerns only women taking up employment and its effect on national income; capability of women vis a vis men is not the issue here. Hence this cannot be assumed.

(B) The argument makes no reference to the type of work women take up, so this is not an assumption.

(C) CORRECT. This is the best choice. Without this assumption the argument would fail.

(D) The conclusion is related only to India, so this cannot be assumed.

Choice (C)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

The British government has confirmed plans to end bursaries for student nurses and midwives from next year. Currently they receive a mixture of a non-means-tested bursary, a means-tested bursary and a reduced-rate student loan to help with their living costs. Replacing bursaries with loans would free up about £800m a year to create additional nursing roles by 2020 and help more students enter the profession, according to the Department of Health. This will allow the cap on student numbers to be lifted, creating up to 10,000 extra training places this parliament. The government claims that this step will end the shortage of nurses and midwives NHS faces now. Which of the following would most weaken the claim above?

- ☐ a) NHS is struggling to fill vacancies now for want of qualified nurses and midwives.
- ☐ b) Nurses and midwives in the UK are not quite well-paid employees.
- ☒ c) Large debts generally deter students from considering a career in nursing and midwifery. (Correct Answer - ✓)
- ☐ d) Student nurses, midwives and allied health professionals currently do not pay tuition fees.

Solution:

The British Government claims that ending bursaries and replacing them with loans would create additional seats for nursing and midwifery students as this step will provide the government with about £800m a year to create additional seats. What information weakens the claim that this measure will end the shortage of nurses NHS faces now? The prediction that more number of students will join is made exclusively on the basis of creation of more seats. But, what if students are unwilling to take the burden of large loans?

- (A) This is a factor responsible for this move, but this will not weaken the claim.
- (B) The very fact that there is a requirement for additional seats suggests that this factor doesn't deter students from joining these courses.
- (C) CORRECT. This information can weaken the claim; students will not join these courses, hence sufficient number of nurses will not be available.
- (D) This is a part of the argument and hence irrelevant to the claim.

Choice (C)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

According to a recent research, the flu virus a person first encounters as a child leaves a permanent "imprint" on the immune system, giving them robust protection against similar strains and much weaker protection against less closely related varieties of the illness. We often find that when seasonal flu strikes, some family members are reduced to shivering wrecks under their duvets, while others get off with little more than a snuffle.

From the passage above, it can be inferred that an effective strategy for protection against seasonal flu in future would be which one of the following?

- ☐ a) Developing two separate seasonal vaccines, one for the young and the other for the old
- ☐ b) Vaccinating every individual in the society against flu
- ☒ c) Developing seasonal vaccines against flu according to the year of birth of a person (Correct Answer - ✓)
- ☐ d) Not vaccinating anyone against flu in the society.

Solution:

Since the virus a person first encounters leaves a permanent mark on the immune system, the person is wellprotected from such strains of virus, but they are not protected against other strains. Similarly, others have little protection against such a strain. What strategy will be the most effective protection against seasonal flu in future? People who encounter a strain of virus different from the one they encountered at first suffer the most when seasonal flu strikes, so it can be inferred that people born in different years were exposed to different strains of virus; hence some suffered while others did not. From this it can be inferred that the best strategy is to tackle people according to the year they were born in.

- (A) Separate vaccines for the young and the old will not be enough to protect all.
- (B) This is quite vague; we have no idea about the protection the vaccination ensures against which strain of virus.
- (C) CORRECT. This strategy can be effective as different strains of flu virus attacked people in different years.
- (D) This is totally irrelevant to protecting people against flu attacks.

Choice (C)

Directions for questions 1 to 15: Select the correct alternative from the given choices.

The Supreme Court has asked search engines like Google, Yahoo and Microsoft to block all pages that carried advertisements or content relating to pre-natal sex determination and asked the government to monitor the sites through a nodal agency. The Court observed that search engines cannot act as a "catalyst" in aggravating the declining sex ratio in India. The search engines contended that they were ready to block advertisements despite it being a major source of revenue, but it was not possible to block texts.

Which of the following can be logically concluded from the passage above?

- ☒ a) Indians will continue to have access to web sites which sold sex determination kits and provided information on the issue. (Correct Answer - ✓)
- ☐ b) Search engines generate revenue only through advertisements.
- ☐ c) Search engines believe that people's right to receive information is sacrosanct.
- ☐ d) Indians have a preference for male child.

Solution:

The Supreme Court has asked search engines to block all pages that carried advertisements or content relating to pre-natal sex determination. The court says that search engines should not act as catalysts in worsening the sex ratio in India. But the search engines say they can block advertisements, but not texts. Based on this situation, what conclusion can be arrived at? The passage states that search engines are ready to block advertisements, but not texts. That means people will still be able to get all information about pre-natal sex determination tests.

- (A) CORRECT. This conclusion can be drawn from the passage as people get all information from the text.
- (B) From 'advertisements being a major source of revenue' in the last sentence, this conclusion cannot be arrived at.
- (C) This is far-fetched. The reasons for not blocking texts could also be technical.
- (D) This statement may be factually correct, but 'declining sex ratio' need not necessarily point to preference for male child, it could also be for female child. Moreover, this point is not the focus of the passage.

Choice (A)